# Yixing Zhao

Curriculum Vitae

Department of Statistics and Actuarial Science, University of Waterloo
Waterloo, Ontario, Canada

y679zhao@uwaterloo.ca | hg-kaibai.github.io | github.com/HG-KaiBai

## Research Interests

Bayesian nonparametric inference; survival analysis under complex censoring; conditional distribution estimation and machine learning; mortality forecasting.

## Education

**University of Waterloo**, Waterloo, Ontario, Canada

*Ph.D. in Statistics (Biostatistics)* Sep 2024 - Present
Supervisor: Prof. Liqun Diao. **Expected completion: 2028.**

*Master of Mathematics in Statistics* Sep 2023 - Aug 2024

*Bachelor of Mathematics in Computational Mathematics* Sep 2020 - Jun 2023
Minors: Statistics; Combinatorics and Optimization; Computing.

## Methodological Research

### MAP-EM Pólya Trees for Survival Analysis with Complex Censoring

- Developed a unified Bayesian nonparametric framework using a Pólya tree prior for right-censored, current-status, and interval-censored event-time data.
- Derived closed-form maximum a posteriori expectation-maximization (MAP-EM) updates; studied the estimator's asymptotic properties.
- Evaluated estimation accuracy in simulation studies: sum of squared errors (SSE) was approximately 50% of that of the nonparametric maximum likelihood estimator and 15% lower than that of fiducial methods.

### Neural Regression Trees for Censored and Uncensored Outcomes

- Developing conditional distribution estimators using a fixed binary-tree structure with covariate-dependent split locations and branch probabilities determined by a neural network.
- The construction yields conditional density, distribution, and survival functions for exact and censored continuous outcomes.
- Research includes validity of the distribution construction and uniform approximation as tree and network complexity increase.

## Conference Presentations

**Zhao, Y.**, and Diao, L. "Neural Regression Trees for Censored and Uncensored Outcomes." Waterloo Student Conference in Statistics, Actuarial Science and Finance, University of Waterloo. October 23, 2026.

**Zhao, Y.**, and Diao, L. "MAP-EM Polya Trees for Survival Analysis with Complex Censoring." 2026 SSC Annual Meeting, Hamilton, Ontario, Canada. June 16, 2026.

## Applied Research Experience

*Research Intern, Mitacs Accelerate & Munich Re* May 2025 - Present

- Developed post-COVID-19 forecasts from historical cause-specific mortality data using DSA variable selection, BMBE modeling, and age-segment selection. Validation SSE was approximately 30% lower than the Lee-Carter benchmark.
- Implemented **mortselect** for reproducible data processing, model fitting, validation, and prediction.
- Built a Bayesian MCMC model using historical mortality shock patterns from World War II, HIV/AIDS, and the Spanish flu for post-pandemic scenario forecasting.

*Research Assistant - Mortality Forecasting* Winter 2024

## Teaching Experience

*Graduate Teaching Assistant, University of Waterloo* Sep 2023 - Present
Provided tutorials, student support, and assessment across nine courses.

| **Course** | **Terms** |
|---|---|
| STAT 431: Generalized Linear Models and their Applications | Winter 2026; Fall 2026 |
| STAT 331: Applied Linear Models | Spring 2026 |
| STAT 231: Statistics | Winter 2026; Spring 2026 |
| STAT 221: Statistics (Non-Specialist Level) | Fall 2025 |
| STAT 332: Sampling and Experimental Design | Fall 2024; Spring 2025 |
| STAT 230: Probability | Spring 2025 |
| STAT 437: Statistical Methods for Life History Analysis | Winter 2025 |
| ME 202: Statistics for Engineers | Winter 2025 |
| STAT 333: Stochastic Processes 1 | Fall 2024 |

## Academic Service and Mentoring

*Mentor, Women in Mathematics Directed Reading Program* Winter 2025; Spring 2026
University of Waterloo
Guided students in reading research papers and textbooks on Bayesian analysis and discussing applications in medicine and finance.

## Scholarships and Awards

| | |
|---|---|
| University of Waterloo Graduate Scholarship | 2025, 2026 |
| Women in Mathematics Mentorship Award | 2025, 2026 |
| Statistics & Actuarial Science Chair's Award | 2024, 2026 |
| International Master's Award of Excellence | 2024 |

## Research Software & Statistical Computing

**mortselect** | R package and source code | Interactive web interface
**Zhao, Y.** (author and maintainer), Diao, L., Meng, Y., and Weng, C.
R package implementing DSA, age-specific band selection, and BMBE forecasts. Developed a browser-based explorer for saved forecasts, with interactive charts and CSV/SVG downloads.
Tools: R, Python, SAS, C++, Git/GitHub.